Model Predictive Control — Spring 2026
Prof. M. Zeilinger

Solution 10
Robust MPC 2

Exercise 1 **Tube-based MPC**

1. Autonomous system is $x^+ = (A + BK)x = 0.8x$

$$\begin{aligned}
\Omega_0 &= [0, 0] \\
\Omega_1 &= [-0.2, 0.1] \\
\Omega_2 &= 0.8 \cdot [-0.2, 0.1] \oplus [-0.2, 0.1] \\
\Omega_3 &= 0.8^2 \cdot [-0.2, 0.1] \oplus 0.8 \cdot [-0.2, 0.1] \oplus [-0.2, 0.1] \\
&\vdots \\
\mathcal{E} &= [-0.2 \cdot \sum 0.8^i, 0.1 \cdot \sum 0.8^i] \\
&= [-0.2 \cdot 5, 0.1 \cdot 5] \\
&= [-1, 0.5]
\end{aligned}$$

2.

$$\begin{aligned}
\tilde{X} &= [-2, 2] \ominus [-1, 0.5] = [-1, 1.5] \\
\tilde{U} &= [-0.8, 0.8] \ominus -0.3 \cdot [-1, 0.5] = [-0.8, 0.8] \ominus [-0.15, 0.3] = [-0.65, 0.5]
\end{aligned}$$

3. The set $\mathcal{X}_f$ must be invariant for the nominal system subject to the tightened constraints.

   Invariance:

$$(A + BK)\mathcal{X}_f = 0.8 \cdot [-1, 1.5] = [-0.8, 1.2] \subset \mathcal{X}_f$$

   Constraint satisfaction:

$$\begin{aligned}
\mathcal{X}_f \subseteq \mathcal{X} \ominus \mathcal{E} &\Leftrightarrow [-1, 1.5] \subseteq [-1, 1.5] \\
K\mathcal{X}_f \subseteq \mathcal{U} \ominus K\mathcal{E} &\Leftrightarrow -0.3 \cdot [-1, 1.5] \subseteq [-0.65, 0.5] \\
&\Leftrightarrow [-0.45, 0.3] \subseteq [-0.65, 0.5]
\end{aligned}$$

Exercise 2 **Nominal MPC with disturbances**

1. See separate MATLAB file for implementation
2. From the simulation we can see that the system converges to a neighborhood of the origin if $\mathcal{W} = [-0.3, 0.3]$. See a state evolution for a specific disturbance realization in Figure 1.
3. However, if we increase the noise it can either not converge or make the problem infeasible.

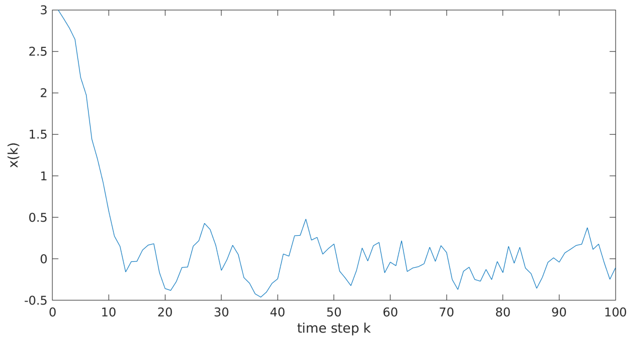

Figure 1: Realization for state evolution with bounded disturbances $\mathcal{W} = [-0.3, 0.3]$.